

CHENGDU SHUANGLIU, China

WMO: 562940

Lat: 30.579N

Lon: 103.947E

Elev: 1625

StdP: 13.85

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.7	35.7	22.7	18.1	46.9	26.8	22.0	45.2	13.1	44.2	11.4	44.6	1.9	350	0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	93.5	76.8	91.6	76.2	89.5	75.3	80.8	88.3	79.4	86.6	78.3	84.9	4.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	159.7	84.8	77.4	151.2	83.5	76.7	148.0	83.0	46.1	88.5	44.6	87.4	43.4	85.5	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	13.1	10.4	8.7	DB	29.5	97.6	1.9	2.2	28.2	99.2	27.1	100.5	26.0	101.7	24.7	103.3
(5)				WB	28.7	83.0	1.9	1.9	27.3	84.4	26.2	85.5	25.2	86.6	23.8	88.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	44.1	48.8	56.4	65.7	72.6	76.9	80.4	80.0	73.3	65.4	56.5	47.0
	DBStd	13.46	4.15	5.41	6.41	6.33	4.67	4.33	3.81	4.50	4.88	4.59	5.19	4.01
	HDD50	396	186	79	14	1	0	0	0	0	0	8	108	
	HDD65	2311	647	454	275	67	4	0	0	1	51	255	557	
	CDD50	5506	4	45	213	471	700	808	941	929	698	476	205	16
	CDD65	1946	0	0	9	87	239	358	476	464	250	62	1	0
	CDH74	15696	0	0	66	619	1650	2839	4537	4372	1429	182	2	0
	CDH80	5591	0	0	6	161	517	958	1745	1758	419	27	0	0
Wind	WSAvg	3.4	2.9	3.3	3.7	3.8	4.0	3.7	3.5	3.5	3.5	3.1	2.8	2.6
Precipitation	PrecAvg	34.9	0.4	0.5	0.9	2.0	2.9	4.6	8.5	8.1	4.6	1.6	0.6	0.3
	PrecMax	53.7	0.8	1.2	2.0	5.2	6.8	9.2	21.7	14.9	9.6	3.0	1.5	0.7
	PrecMin	25.2	0.0	0.0	0.1	0.4	0.8	1.1	3.0	3.3	1.3	0.6	0.1	0.0
	PrecStd	6.5	0.2	0.4	0.5	0.9	1.3	2.1	4.1	3.0	2.4	0.7	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	68.4	80.3	88.3	92.8	93.5	96.5	97.1	92.9	83.9	73.4	61.1
		MCWB	48.8	54.1	61.8	67.0	69.1	73.3	78.4	77.4	76.5	67.8	61.2	50.6
	2%	DB	56.9	64.4	75.2	84.4	88.2	91.0	93.3	93.6	88.1	78.8	69.7	57.5
		MCWB	47.1	52.6	59.4	65.9	67.9	73.0	77.7	78.1	74.3	66.6	59.9	49.1
	5%	DB	53.6	60.8	71.4	80.8	85.8	88.0	91.3	91.5	84.5	75.2	66.5	55.5
		MCWB	46.0	51.0	58.1	65.2	68.2	72.8	77.4	77.7	73.0	65.3	58.3	48.4
	10%	DB	51.3	57.5	67.7	77.1	82.5	85.8	88.6	89.2	81.0	72.0	64.2	53.6
		MCWB	44.8	49.8	56.9	63.9	67.2	73.0	76.9	76.9	71.8	64.8	57.8	47.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	55.6	64.8	71.9	74.6	80.0	83.2	82.3	79.8	72.1	63.2	53.6
		MCDB	57.3	65.2	75.2	83.2	83.6	86.4	90.8	90.3	88.2	77.9	70.0	57.8
	2%	WB	48.6	53.8	61.5	68.9	72.4	77.8	81.1	80.6	77.0	69.0	61.5	51.7
		MCDB	54.2	61.3	70.3	78.4	80.7	84.0	88.2	89.0	83.8	74.3	66.9	55.6
	5%	WB	47.2	52.2	59.3	67.0	70.9	76.4	79.7	79.3	75.0	67.4	60.2	50.2
		MCDB	52.0	59.4	68.0	76.7	79.1	82.3	86.4	87.1	81.6	72.6	64.5	53.9
	10%	WB	45.7	50.3	57.6	65.1	69.6	75.0	78.6	78.2	73.3	66.0	58.9	48.8
		MCDB	49.9	56.5	65.9	74.7	78.2	81.3	84.8	85.4	79.3	70.7	62.5	52.1
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.7	13.8	15.7	15.6	13.4	13.5	14.0	11.5	10.2	10.2	10.1
		MCDBR	16.4	18.4	21.3	22.5	22.1	18.8	18.7	18.4	17.5	15.9	15.2	14.6
		MCWBR	8.8	8.8	8.9	8.1	7.0	6.6	7.0	6.6	6.7	6.3	7.1	8.0
	5% WB	MCDBR	14.5	16.6	18.5	19.0	16.3	14.4	15.5	15.8	14.9	12.8	12.4	11.7
		MCWBR	8.6	8.5	8.6	8.2	6.9	7.1	7.4	7.0	7.0	6.1	6.5	7.0
Clear-Sky Solar Irradiance	taub	0.680	0.722	0.756	0.743	0.704	0.673	0.700	0.730	0.746	0.753	0.733	0.690	
	taud	1.475	1.432	1.415	1.476	1.585	1.672	1.633	1.564	1.516	1.471	1.460	1.492	
	Ebn at Noon	168	178	186	198	207	213	207	198	186	169	154	156	
	Edh at Noon	80	91	98	95	86	79	82	86	88	86	81	75	
All-Sky Solar Radiation	RadAvg	626	760	994	1249	1341	1261	1350	1296	927	736	644	589	
	RadStd	82	115	106	125	157	156	192	177	110	101	51	84	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.01	+0.90	-2.21	+1.78	N/A	N/A	-90	-164	+295	+175
(47)	Regional (6 neighbors)	+0.50	N/A	-1.15	+0.74	N/A	N/A	-90	-164	+295	+175

Nomenclature: See separate page